FRAUD PREVENTION & DETECTION TECHNOLOGIES

CYS339

DATA ANALYIS

UNIT 1

DATA ANALYSIS

- **Data** are facts and statistics collected together for reference or analysis.
- **Analysis** is a detailed examination of the elements or structure of something.
- **Data analysis** is the process of examining raw data to draw conclusions about the information it contains, often using statistical techniques, to understand patterns, trends, and relationships.

FRAUD

- **Fraud** refers to the deliberate use of deception, dishonesty, or false representation to gain an unfair advantage, often resulting in financial or personal loss to the victim.
- In the context of cybersecurity, fraud typically involves unauthorized access to systems, data theft, or manipulation of information for malicious purposes.
- Examples include **identity theft**, **phishing**, **credit card fraud**, and **insider threats**.

DETECTION TECHNOLOGIES

- **Detection technologies** are tools, systems, or methodologies used to identify and respond to malicious activities, anomalies, or threats within a network or system.
- These technologies are designed to monitor, analyze, and flag suspicious behavior, enabling organizations to mitigate risks and prevent cyber attacks.
- Examples include **Intrusion Detection Systems (IDS), firewalls, machine learning algorithms**, and **anomaly detection systems.**

THREE REASONS DATA ANALYSIS IS MUST

1. **Improved Efficiency in Detecting and Preventing Cyber Threats**

- Data analysis **enhances the efficiency** of cybersecurity operations by automating the detection of anomalies and potential threats.
- Advanced analytical tools, such as **Intrusion Detection Systems (IDS)** and **Security Information and Event Management (SIEM)** systems, process vast amounts of data in real-time to identify suspicious behavior.
- This allows cybersecurity teams to respond quickly and effectively, minimizing downtime and preventing breaches

THREE REASONS DATA ANALYSIS IS MUST (CONTINUES…)

2. **Enhanced Understanding of Attack Patterns and Vulnerabilities**

- Data analysis helps cybersecurity professionals **gain deeper insights** into attack patterns, tactics, and techniques used by malicious actors.
- By analyzing historical attack data and threat intelligence, organizations can identify vulnerabilities in their systems and develop strategies to strengthen their defenses.
- This proactive approach reduces the likelihood of successful cyberattacks and improves overall security posture.

THREE REASONS DATA ANALYSIS IS MUST (CONTINUES…)

3. **Informed Decision-Making for Threat Response**

- Data analysis is critical in cybersecurity for making informed decisions about how to respond to threats and vulnerabilities.
- By analyzing data from logs, network traffic, and security incidents, cybersecurity professionals can identify patterns of malicious activity, assess risks, and prioritize responses.
- This enables organizations to take proactive measures to mitigate threats and reduce the impact of cyberattacks.

DATA ANALYSIS IN AUDITING

- A cybersecurity **audit** is a process that includes the analysis of a company's IT infrastructure for **review** and improvement.

❑ **Purpose**

- Data analysis in **auditing** is primarily used to assess the **accuracy, completeness,** and **reliability** of financial records and transactions.

❑ **Scope**

- Focuses on **evaluating internal controls, compliance with regulations,** and the **overall financial health** of an organization.

DATA ANALYSIS IN AUDITING (CONTINUES…)

❑ **Tools and Techniques**

- Uses tools like **Computer-Assisted Audit Techniques (CAATs)** and software such as **ACL** or **IDEA**.
- Techniques include **trend analysis, ratio analysis,** and **variance analysis.**

❑ **Outcome**

- **Provides assurance** to stakeholders (e.g., shareholders, regulators) that financial statements are free from material misstatements.

DATA ANALYSIS IN INVESTIGATION

- **Investigation** is the act or process of **examining** a crime, problem, statement, etc.

❑**Purpose**

- Data analysis in investigations is **used to uncover evidence** of fraud, misconduct, or illegal activities.
- It aims to **identify the root cause** of issues and determine accountability.

❑ **Scope**

- **Focuses on specific incidents** or allegations, often requiring a deep dive into detailed transactional data.
- Involves **examining all relevant data** rather than relying on sampling.

DATA ANALYSIS IN INVESTIGATION (CONTINUES…)

❑ **Tools and Techniques**

- Uses forensic data analysis tools like **EnCase**, **FTK**, or **Tableau** for visualization.
- Techniques include **link analysis**, **pattern recognition**, and **anomaly detection**.

❑ **Outcome**

- **Provides actionable evidence for legal proceedings**, disciplinary actions, or recovery of losses.

KEY DIFFERENCES BETWEEN AUDITING AND INVESTIGATION

1. **Objective**

- **Auditing** focuses on providing assurance and compliance.
- **Investigation** focuses on uncovering wrongdoing and gathering evidence.

2. **Approach**

- **Auditing** is systematic and often follows a predefined plan.
- **Investigation** is reactive and driven by specific allegations or incidents.

KEY DIFFERENCES BETWEEN AUDITING AND INVESTIGATION (CONTINUES…)

3. **Data Depth**

- **Auditing** typically uses sampling and high-level analysis.
- **Investigation** requires exhaustive examination of detailed data.

4. **Outcome Use**

- **Auditing** results are used for stakeholder assurance and regulatory compliance.
- **Investigation** results are used for legal or disciplinary actions.

WHAT CAN DATA ANALYSIS DO?

1. Identify Patterns and Trends in Cyber Threats
2. Support Decision-Making for Incident Response
3. Improve Efficiency in Threat Detection
4. Detect Anomalies and Potential Breaches
5. Enhance Threat Intelligence and Proactive Defense

WHAT DATA ANALYSIS CANNOT DO IN CYBERSECURITY

1. **Replace Human Judgment in Incident Response**

- A system may flag a legitimate user's unusual activity as suspicious, requiring human intervention to verify the alert.

2. **Guarantee Accuracy in Threat Detection**

- False positives and false negatives can occur if an Intrusion Detection System (IDS) is not well configured.

COMMON ANALYSIS PROCEDURES & FUNCTIONS

UNIT 2

COMMON ANALYSIS PROCEDURES

FINANCIAL RATIO, VARIANCE, PERCENTAGE, AGING & AVERAGE

FINANCIAL RATIO

- **Ratio** is the quantitative relation between two amounts showing the number of times one value contains or is contained within the other.
- A **ratio** compares values e.g., 3 to 1 (3:1).
- **Financial ratios** are created with the use of numerical values taken from financial statements to gain meaningful information about a company.

FINANCIAL RATIO (CONTINUES…)

- These ratios help stakeholders (investors, creditors, management) assess:

1. **Liquidity** (ability to meet short-term obligations)
2. **Profitability** (ability to generate earnings)
3. **Leverage** (degree of debt reliance)
4. **Efficiency** (effectiveness in asset utilization)

- In Cybersecurity it helps the CISO to make the above analysis.

FINANCIAL RATIO (CONTINUES…)

- **Advantages of using Financial Ratio**

1. **Simplifies Complex Data** - Ratios condense large datasets into comparable metrics, making trends easier to identify.
2. **Cost-Benefit Analysis** - Helps quantify the financial impact of security investments.
3. **Prioritization** – Identifies high-risk areas by correlating security incidents with financial losses.

FINANCIAL RATIO (CONTINUES…)

❑ **Limitations of Financial Ratio**

1. **Oversimplification** – May ignore qualitative factors (e.g., employee awareness, emerging threats).
2. **Data Accuracy Issues**
3. **Lagging Indicators** - Many ratios reflect past performance rather than predicting future threats.

FINANCIAL RATIO (CONTINUES…)

❑ **Question**

1. A company spends $50,000 annually on cyber security measures. The total financial loss due to cyber attacks before implementing the measures was $200,000 and after implementation the loss reduced to $80,000 Dollars.

i. Calculate the Return on Investment.

ROI = (Total Benefit - Cost of Investment) / Cost of Investment * 100%

= [(200000 – 80000) - 50000/ 50000] * 100%

= [(120000 – 50000)/ 50000] * 100%

= (70000 / 50000) * 100%

= 1.4 * 100%

= 140%

This means that for every dollar spent on cybersecurity, the company gained $1.40 in reduced financial losses.

VARIANCE

- **Variance** measures the differences between **expected** and **actual** value to identify deviations.
- **Deviation** is a measure which is used to find the difference between the observed value and the expected value of a variable.
- It can be applied in;

1. **Financial Auditing**
2. **Cybersecurity**

VARIANCE (CONTINUES…)

- **Types of Variance**

1. **Favourable Variance** – Actual < Expected (e.g., lower costs)
2. **Unfavourable Variance** – Actual > Expected (e.g., higher breach costs)

Variance = Actual − Expected

VARIANCE (CONTINUES…)

❑ **Question**

1. A company records an expected daily login count of 5000 for each employee. On a particular day, the actual recorded log in count was 7500.

i. Calculate the variance.

ii. Determine whether it's favorable or unfavorable.

Variance = Actual – Expected

= 7500 – 5000

= 2500

Favorable Variance = Actual < Expected

Unfavorable Varia.. = Actual > Expected

Therefore, the **Variance is Unfavorable**.

NOTE: + Variance is Unfavorable

– Variance is Favorable

VARIANCE (CONTINUES…)

- **Advantages of using Variance**

1. **Anomaly Detection** – Helps identify deviations from normal behavior (e.g., unusual network traffic, login attempts).
2. **Supports Trend Analysis** – Reveals changes over time (e.g., increasing attack frequency).
3. **Useful in Statistical Models** – Key in machine learning and predictive analytics for cybersecurity (e.g., outlier detection).

VARIANCE (CONTINUES…)

- **Limitations of using Variance**

1. **Sensitive to Outliers** – Leading to false positives in threat detection.
2. **May Miss Subtle Attacks** - Low variance attacks (e.g., slow, persistent threats) can evade detection.

PERCENTAGE

- In mathematics, a **percentage** is a number or ratio expressed as a fraction of 100.
- It is often denoted using the percent sign (%).
- It is also applied in;

1. **Financial Analysis**
2. **Cybersecurity**

PERCENTAGE (CONTINUES…)

- **Common Percentage Formulas**

1. **Percentage of Total**

   Percentage = (Part / Whole) * 100%

2. **Percentage Change**

   Percentage Change = [(New – Old) / Old] * 100%

PERCENTAGE (CONTINUES…)

❑ **Question**

1. A cybersecurity team analyzes 500 phishing attempts in a month. Out of these, 35 where successful.

i. Calculate the attack success rate percentage.

Percentage of Total = (Part / Whole) * 100%

= (35 / 500) * 100%

= 7%

Therefore, the attack success rate was 7%

PERCENTAGE (CONTINUES…)

- **Advantages of Percentages**

1. **Simplifies Interpretation**
2. **Trend Identification** – Helps track changes over time (e.g., "Phishing attempts increased by 15% this quarter").
3. **Normalizes Data** – Allows comparison across different scales.

PERCENTAGE (CONTINUES…)

❑ **Limitations of using Percentages**

1. **Limited in Isolation** – Percentages alone don't explain causality or risk impact (requires additional metrics).
2. **Potential for Misleading Context** – Without proper reference points, percentages can be misinterpreted (e.g., "50% of attacks" without knowing total volume).
3. **Can Mask Absolute Values** – A small percentage change may represent a critical threat if the base number is large (e.g., "1% increase in breaches" could mean thousands of incidents).

AGING

- **Aging** analysis examines the time elapsed since an event occurred to detect anomalies that may indicate fraud.
- It can be applied in Cybersecurity for;

1. **Time-Based Segmentation** – Groups data by age (e.g., inactive accounts suddenly accessed after 6+ months)
2. **Thresholds & Rules** – Flags unusual delays (e.g., late invoice submissions)
3. **Trend Analysis** – Detects abnormal aging patterns (e.g., old credentials reused in a brute-force attack)

AGING (CONTINUES…)

❑ **Question**

1. A company tracks unpatched vulnerabilities overtime. The data is as follows:

Age (Days)	Unpatched Vulnerabilities
0 – 30	25
31 – 60	20
61 – 90	10
91+	5

i. What percentage of vulnerabilities are odder than 60 days?

Percentage of Total = (Part / Whole) * 100%

= [(10 + 5) / (25 + 20 + 10 + 5)] * 100%

= (15 / 60) * 100%

= 0.25 * 100%

= 25%

AGING (CONTINUES…)

ADVANTAGES

1. **Simple & Scalable**
2. **Detects Timing Anomalies**

LIMITATIONS

1. **False Positives**
2. **Static Thresholds** – Requires manual updates to adapt to new fraud tactics.

AVERAGE (MEAN) ANALYSIS

- **Average** measures central tendency to identify typical values.
- It is applied in;

1. **Finance**
2. **Cybersecurity**

- Formula is;

$$\textbf{Average} = \sum \textbf{Values} / \textbf{Count}$$

AVERAGE (MEAN) ANALYSIS (CONTINUES…)

❑ **Question**

1. A security operations center measures the mean time to detect cyber threats over 5 incidents. Detection Times (hours) : [4,5,3, 6,7]

i. Calculate the mean time

Average = ∑ Values / Count

= (4 + 5 + 3 + 6 + 7) / 5

= 25 / 5

= 5

Therefore, the mean time to detect the Cyberthreats is 5 hours.

AVERAGE (MEAN) ANALYSIS (CONTINUES…)

ADVANTAGES

1. **Simple & Fast** – Easy to compute and interpret.
2. **Baseline Comparison** – Helps distinguish normal vs. suspicious behavior.

LIMITATIONS

1. **False Positives/Negatives**
2. **Static Baseline Issue** – Requires updates to reflect evolving behavior.

COMMON ANALYSIS FUNCTIONS

FILTER, SORT, JOIN, APPEND, PIVOT, TREND, REGRESSION, RATIO, VLOOKUP, INDEX/MATCH, XLOOKUP, HLOOKUP, EYELOOKUP, GROUPING, SUMMARIZATION, AND STRATIFICATION

FILTER

- **Filtering** refers to removing unwanted material.
- The **FILTER** function in Excel is used to return specific data from a range based on one or more conditions.
- It helps extract relevant data dynamically without manually sorting or searching.
- It can be applied in Cybersecurity to;

1. **Identifying Malicious IPs Quickly**
2. **Tracking Phishing Attacks**

FILTER (CONTINUES…)

❑ **Syntax of The FILTER Function**

=FILTER(array, include, [if_empty])

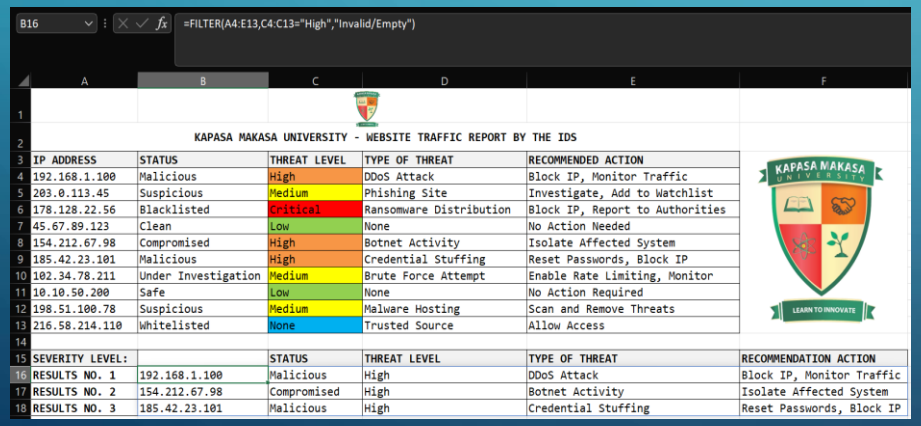
- **array** ⟶ The range of data you want to filter.
- **include** ⟶ The condition(s) used to filter the data.
- **if_empty** (optional) ⟶ The value to return if no data matches the condition.

FILTER (CONTINUES…)

SINGLE CONDITION

=FILTER(A4:E13,C4:C13="High", “Invalid/Empty")

- This will return only rows where Threat Level (column C) is "High“.



KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS

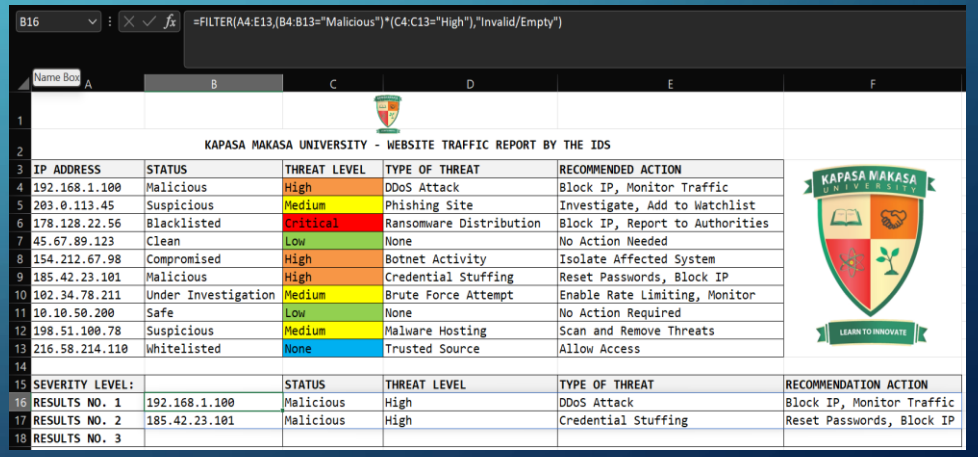
IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION
192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist
178.128.22.56	Blacklisted	Critical	Ransomware Distribution	Block IP, Report to Authorities
45.67.89.123	Clean	Low	None	No Action Needed
154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP
102.34.78.211	Under Investigation	Medium	Brute Force Attempt	Enable Rate Limiting, Monitor
10.10.50.200	Safe	Low	None	No Action Required
198.51.100.78	Suspicious	Medium	Malware Hosting	Scan and Remove Threats
216.58.214.110	Whitelisted	None	Trusted Source	Allow Access

SEVERITY LEVEL:		STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
RESULTS NO. 1	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
RESULTS NO. 2	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
RESULTS NO. 3	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP

MULTIPLE CONDITIONS

=FILTER(A4:E13, (B4:B13="Malicious") * (C4:C13="High"), "Invalid/Empty")

- The * acts as **AND** logic, meaning both conditions must be met.



KAPASA MAKASA UNIVERSITY - WEBSITE TRAFFIC REPORT BY THE IDS

IP ADDRESS	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDED ACTION
192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
203.0.113.45	Suspicious	Medium	Phishing Site	Investigate, Add to Watchlist
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SEVERITY LEVEL:		STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
RESULTS NO. 1	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
RESULTS NO. 2	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP
RESULTS NO. 3					

FILTER (CONTINUES…)

- **Filter Data Dynamically**

  =FILTER(A4:E13,C4:C13=B15,"Invalid/Empty")

- This will update results automatically as the value in **B15** changes.

15	SEVERITY LEVEL:	High	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	192.168.1.100	Malicious	High	DDoS Attack	Block IP, Monitor Traffic
17	RESULTS NO. 2	154.212.67.98	Compromised	High	Botnet Activity	Isolate Affected System
18	RESULTS NO. 3	185.42.23.101	Malicious	High	Credential Stuffing	Reset Passwords, Block IP

15	SEVERITY LEVEL:	None	STATUS	THREAT LEVEL	TYPE OF THREAT	RECOMMENDATION ACTION
16	RESULTS NO. 1	216.58.214.110	Whitelisted	None	Trusted Source	Allow Access
17	RESULTS NO. 2					
18	RESULTS NO. 3					

FILTER (CONTINUES…)

ADVANTAGES

1. **Reduced Noise** – Eliminates irrelevant data.
2. **Faster Processing** – Decreases data volume.

LIMITATIONS

1. **Risk of Over-Filtering**
2. **Potential Data Loss**

SORT

- Organize data in meaningful order.
- Arranges data in ascending or descending order.
- The **SORT** Function syntax in Excel is;

=SORT(array,[sort_index],[sort_order],[by_col])

=SORTBY(array, by_array1, [sort_order1], [by_array2, sort_order2],…)

SORT (CONTINUES…)

	A	B	C	D
1	Common Vulnerability Scoring Systems (CVSS)			
2	RANGE		STATUS	
3	9.0 - 10.0		Critical	
4	0.0 - 0.0		Non	
5	4.0 - 6.9		Medium	
6	0.1 - 3.9		Low	
7	7.0 - 8.9		High	

F2 =SORT(A3:D7)

	A	B	C	D	E	F	G	H	I
1	Common Vulnerability Scoring Systems (CVSS)								
2	RANGE		STATUS			0.0 - 0.0	0	Non	0
3	9.0 - 10.0		Critical			0.1 - 3.9	0	Low	0
4	0.0 - 0.0		Non			4.0 - 6.9	0	Medium	0
5	4.0 - 6.9		Medium			7.0 - 8.9	0	High	0
6	0.1 - 3.9		Low			9.0 - 10.0	0	Critical	0
7	7.0 - 8.9		High						

SORT (CONTINUES…)

ADVANTAGES

1. **Prioritization of Threats** – Sorts data by severity (e.g., high-risk alerts first), enabling faster response to critical incidents.

LIMITATIONS

1. **Context Loss** – Sorting by one column may hide relationships in other fields.
2. **Misleading Order** – Ascending/descending sorts may unintentionally emphasize less relevant data.

JOIN

- **Join** combines related data from different sources (**tables** or **databases**).
- Combining is usually based on a **common key** (in databases we use a **PRIMARY KEY** and **FOREIGN KEY**).
- Common types of joins include:

i. **Inner join**

ii. **Right join**

iii. **Full outer join**

iv. **Left join**

- In Excel the syntax for **TEXTJOIN** Function is;

=TEXTJOIN(delimiter, ignore_empty, text1, [text2], …)

JOIN (CONTNUES…)

A9 =TEXTJOIN(",",TRUE,A3:D7)

	A	B	C	D	E	F	G	H	I
1	Common Vulnerability Scoring Systems (CVSS)								
2	RANGE		STATUS						
3	9.0 - 10.0		Critical						
4	0.0 - 0.0		Non						
5	4.0 - 6.9		Medium						
6	0.1 - 3.9		Low						
7	7.0 - 8.9		High						
8									
9	9.0 - 10.0,Critical,0.0 - 0.0,Non,4.0 - 6.9,Medium,0.1 - 3.9,Low,7.0 - 8.9,High								

JOIN (CONTNUES…)

- Bellow is how to use **INNER JOIN** in a relational database.

```
CREATE TABLE students (
   SIN INT PRIMARY KEY,
   [FIRST NAME] VARCHAR(10),
   [LAST NAME] VARCHAR(10),
   GENDER VARCHAR(6) NOT NULL
);
```

```
CREATE TABLE results (
   SIN INT PRIMARY KEY,
   [COURSE CODE] VARCHAR(10),
   [CA /40] FLOAT,
   [TOTAL /100]  FLOAT,
   GRADE VARCHAR(2) NOT NULL,
   GENDER VARCHAR(6) NOT NULL,
   FOREIGN KEY (SIN) REFERENCES students(SIN)
);
```

```
INSERT INTO students
VALUES (20231392, "James", "Soko", "Male");
```

```
INSERT INTO results
VALUES (20231392, "CYS339", 33, 47, "A", "Male");
```

```
ALTER TABLE results
DROP GENDER;
```

```
SELECT *
FROM students
INNER JOIN results ON students.SIN = results.SIN
WHERE students.SIN = 20231392;
```

JOIN (CONTINUES…)

- **Advantages**

1. **Enhanced Correlation** – Combines data from multiple sources (e.g., firewall logs + authentication events) to detect complex attack patterns.
2. **Comprehensive Threat Detection** – Links related events (e.g., matching IPs in network logs with known threat intelligence feeds).

JOIN (CONTINUES…)

- **Limitations**

1. False Positives/Negatives
2. Complexity
3. Privacy Risks

APPEND

- **Append** involves **adding** new rows of data to an existing database.
- Usually when both have the same structure.

```
INSERT INTO students
VALUES (20231392, "James", "Soko", "Male");
```

SIN	FIRST NAME	LAST NAME	GENDER
20231392	James	Soko	Male

APPEND (CONTINUES…)

- **Advantages**

1. **Data Aggregation** – Combines multiple datasets.
2. **Simplified Historical Analysis** - Enables trend detection by appending new data to historical records (e.g., tracking attack patterns over months).
3. **Compatibility with SIEMs** – Many security tools (e.g., Splunk, ELK) rely on appended logs for correlation and alerting.

APPEND (CONTINUES…)

- **Limitations**

1. Data Duplication
2. Increased Storage Costs
3. Performance Overhead

JOIN & APPEND

❑ Question

1. Join and append are two (2) prevalent data analysis functions. Highlight the key similarities and differences between these two functions.

JOIN & APPEND (CONTINUES…)

Similarities	Differences
Both are used to combine data.	**Join** combines two tables using a common column while **append** adds new rows to an existing table.
Both work with tables (or lists).	
Both help organize and analyze data.	

PIVOT

- **Pivot** summarize and cross-tabulate data.
- Transforms data by rearranging rows and columns to summarize key information.
- The **GETPIVOTDATA** function is used to extract specific data from a PivotTable dynamically.

GETPIVOTDATA(data_field, pivot_table, [field1, item1, field2, item2, ...])

PIVOT (CONTINUES…)

- **data_field** – The name of the field you want to retrieve (in quotes).
- **pivot_table** – A cell reference to any part of the PivotTable.
- **field1, item1,** etc. – Optional pairs that specify which field and value to filter by.
- For example;

=GETPIVOTDATA("Total Sales", A1, "Salesperson", "Alice")

PIVOT (CONTINUES…)

PIVOT (CONTINUES…)

ADVANTAGES

1. **Enhanced Data Summarization**
2. **Quick Pattern Recognition**

LIMITATIONS

1. **Over-Simplification Risk**
2. **Static Output** – Requires manual updates if source data changes, unlike real-time analytics.

TREND

- **Trend** helps to identify patterns over time.
- Helps in forecasting and decision making.
- The **TREND** function in Excel predicts future values based on existing data using linear regression.

=TREND(known_y's, [known_x's], [new_x's], [const])

TREND (CONTINUES…)

- **known_y's** – The dependent values (the values you want to predict from).
- **known_x's** (optional) – The independent values (the existing input data).
- **new_x's** (optional) – The values for which you want to predict new y values.
- **const** (optional, default = TRUE) – If **TRUE**, the equation includes an intercept; if **FALSE**, the regression line passes through zero.

TREND (CONTINUES…)

ADVANTAGES

1. **Anomaly Detection**
2. Helps **predict future attacks** by analyzing historical trends

LIMITATIONS

1. **False Positives/Negatives**
2. **Historical Bias**
3. **Requires large datasets for accuracy**
4. **Time Lag –** Trends reflect past events

REGRESSION

- **Regression** model relationships between variables.
- Common types:

i. Linear regression

ii. Logistic regression.

REGRESSION (CONTINUES…)

ADVANTAGES

1. **Predictive Capability** – Forecasts future threats.
2. **Anomaly Detection** – Identifies outliers.
3. **Root Cause Analysis**
4. **Automation Potential** – Integrates with AI/ML for real-time threat scoring.

LIMITATIONS

1. Lagging Indicators
2. Data Quality Dependence

REGRESSION (CONTINUES…)

❑ **Question**

1. A companies revenue depends on the amount spent on advertising, the relationship between sales and the advertising spent is modeled by the following regression equation.

   Sales = $500 + (10 * Advertising)

i. If a company spends $2000 on advertising, predict the expected sales revenue.

Sales = $500 + (10 * Advertising)

= $500 + (10 * $2000)

= $500 + $20000

= $20500

VLOOKUP

- **Vertical Lookup** is an excel function.
- It is used to search for a specific value in a column and return a corresponding value from another column in the same row.
- **Key Characteristics of VLOOKUP –**

1. **Searches vertically** (top to bottom) in the first column of a table.
2. Can perform **exact** (FALSE) or **approximate** (TRUE) matches.

VLOOKUP (CONTINUES…)

- basic **syntax –**

  =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

1. **lookup_value –** the value you want to search for.
2. **table_array –** the range of cells containing the data.
3. **col_index_num –** the column number from which to retrieve the value.
4. **range_lookup –** default is TRUE (1) used for an approximate vale and FALSE (0) for an exact match.

VLOOKUP (CONTINUES…)

LIMITATIONS OF VLOOKUP

1. Left-to-Right Lookup Only
2. Exact Match Requirement
3. Slower with Large Datasets

ALTERNATIVES TO VLOOKUP

1. INDEX-MATCH
2. XLOOKUP
3. HLOOKUP

GROUPING

- **Grouping** is the process of organizing data into **categories** or **clusters** based on shared characteristics.
- It enables analysts to:

1. Identify patterns and trends
2. Simplify complex datasets
3. Perform aggregate calculations
4. Compare subsets of data

GROUPING (CONTINUES…)

- Key **Characteristics** of grouping include –

1. **Categorization** – data is divided into meaningful buckets.
2. **Aggregation** – enables summary statistics per group.
3. **Comparative Analysis** – allows group-to-group comparisons.
4. **Dimensionality Reduction** – simplifies large datasets.

GROUPING (CONTINUES…)

- **Methods of Grouping** include –

1. **Basic Grouping Techniques**

i. **Manual Categorization** – Creating custom groups based on domain knowledge. For example, grouping alerts as "Malicious".

ii. **Automated Grouping** – Using functions like GROUP BY (SQL) or PivotTables (Excel).

GROUPING (CONTINUES…)

2. **Advanced Grouping Methods**

i. **Binning** – Creating ranges for continuous data (e.g., **CVSS**).

ii. **Clustering** – **Machine learning** approach for finding natural groupings (e.g., clustering malware samples by behavior using algorithms like **K-means clustering**).

GROUPING (CONTINUES…)

- Cybersecurity **Applications of Grouping** include –

1. Threat Intelligence Analysis
2. SIEM Alert Management
3. Vulnerability Management
4. Incident Investigation

GROUPING (CONTINUES…)

ADVANTAGES

1. Pattern Recognition
2. Noise Reduction
3. Simplifies large datasets by categorizing data
4. Threat Correlation
5. Prioritization

DISADVANTAGES

1. May hide important outliers
2. Incorrect grouping can link unrelated events
3. Static Grouping

SUMMARIZATION

- Is the process of **distilling large datasets into concise**, meaningful statistics and metrics that represent key characteristics of the data.
- Its **purpose** is to –

i. Reduce data complexity

ii. Support decision-making

SUMMARIZATION (CONTINUES…)

Summarization Techniques

1. **Descriptive Statistics** – include count, median, mean, mode, Standard Deviation (SD) and range.
2. **Frequency Distributions** – tally of how often values occur.
3. **Aggregate Functions** – common functions include **SUM()**, **AVG()**, **COUNT()**, **MAX()**, **MIN()**, and **PERCENTILE()**.

SUMMARIZATION (CONTINUES…)

VISUALIZATION OF SUMMARIES

1. Bar Charts
2. Pie Charts

ADVANTAGES

1. Simplifies Complex Data
2. Improves Decision-Making
3. Enhances Reporting

STRATIFICATION

- Is the process of **dividing a population into homogeneous subgroups (strata)** based on shared characteristics.
- **Characteristics of stratification** include –

1. **Segmentation** – creates meaningful **subgroups**.
2. **Homogeneity** – members within **strata are similar**.
3. **Heterogeneity – strata differ** significantly from each other.
4. **Representation** – maintains population **proportions**.

STRATIFICATION (CONTINUES…)

ADVANTAGES

1. Targeted Threat Detection
2. Reveals attack patterns
3. Organizes data according

DISADVANTAGES

1. Over-Segmentation Risk
2. Increased Complexity
3. Stratification Bias

DATA ANALYSIS

UNIT 3

TYPE OF DATA ANALYSIS

1. Quantitative

- It involves **numerical data** and **statistical techniques** to detect, analyze, and prevent cyber threats.
- It deals with data which is **numerical, measurable** and **comparable.**
- Understanding the **types of quantitative data** analysis will use the key word **$(SDP)^2$ CT** which includes **Statistical, Spatial, Descriptive, Diagnostic, Prescription, Predictive, Comparative** and **Time Series Analysis.**

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis

A. Descriptive Analysis

- **Summarizes** and **describes key features** of security data.
- **Cybersecurity Application** – summarizing DDoS attack durations.
- **Tools – SIEM** dashboards, **Excel** and **Tableau**.

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

B. Diagnostic Analysis

- Examines **why** security events occurred.
- **Cybersecurity Application** – Determining **root causes** of breaches.
- **Tools** – Splunk.

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

C. Predictive Analysis

- Uses historical data to **forecast future threats**.
- **Cybersecurity Applications – Predicting attack volumes** based on historical trends and **Forecasting potential DDoS attack times.**
- **Tools** – Machine Learning (ML).

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

D. Prescriptive Analysis

- **Recommends** actions based on data insights.
- **Cybersecurity Applications – automated patching** prioritization and dynamic firewall rule recommendations.
- **Tools** – SOAR platforms.

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

E. Comparative Analysis

- **Benchmarks** security metrics against standards.
- **Cybersecurity Application** – measuring security posture against industry benchmarks.
- **Tools** – Security Scorecard.

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

F. Time Series Analysis

- Examines security data points **over time**.
- **Cybersecurity Application** – detecting seasonal attack patterns.
- **Tools – Elasticsearch**, and **Pandas**.

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

G. Statistical Analysis

- Applies **mathematical models** to security data.
- **Cybersecurity Applications – calculating probabilities** of breach scenarios and performing **regression** analysis on attack vectors.
- **Tools – R** and **Python** (NumPy and SciPy)

TYPE OF DATA ANALYSIS (CONTINUES…)

Types of Quantitative Data Analysis (Continues…)

H. Spatial Analysis

- Examines **geographic** patterns in threats.
- **Cybersecurity Application** – correlating physical locations with breach attempts.
- **Tools – GeoIP** and **ArcGIS**.

TYPE OF DATA ANALYSIS (CONTINUES…)

Other types of Quantitative Data Analysis

i. Sorting

- **Organizing data in a structured order** (e.g., chronological, severity-based) to identify trends, outliers, or critical events.
- **Tools – Splunk, Nessus** and **Qualys**.
- **Strength –** identifying high-priority threats.
- **Limitation** - false positives.

TYPE OF DATA ANALYSIS (CONTINUES…)

Other types of Quantitative Data Analysis (Continues…)

ii. Categorization

- **Grouping data** into predefined classes (e.g., threat types, user roles) for structured analysis.
- **Tools – VirusTotal** and **Wireshark**.
- **Strength –** simplifies reporting.
- **Limitation** - misclassification leads to flawed policies.

TYPE OF DATA ANALYSIS (CONTINUES…)

Other types of Quantitative Data Analysis (Continues…)

iii. Relationship

- Causal **links** between events (**correlation**).
- **Tool – Splunk.**
- **Strength** – grouping related events.
- **Limitation** - false correlations.

TYPE OF DATA ANALYSIS (CONTINUES…)

Other types of Quantitative Data Analysis (Continues…)

iv. Profiling (Behavioural Analysis)

- Establishing **baselines of normal behavior** to detect anomalies (e.g., unusual login times).
- **Tools – Cisco Stealthwatch** and **Splunk**.
- **Strength –** adapts to evolving threats.
- **Limitation** - high false positives.

TYPE OF DATA ANALYSIS (CONTINUES…)

2. Qualitative

- **focuses on non-numerical data** (e.g., text, logs, narratives) to understand meaning, patterns, and motivations behind cyber threats.
- It interprets **why** and **how** an attack occurred.
- There are two major **types of qualitative data analysis**, which is **behavioral** and **contextual** analysis.

TYPE OF DATA ANALYSIS (CONTINUES…)

i. **Contextual**

- Answers **"Why did this happen?".**
- **Tool – Splunk.**
- **Strength – deepens** understanding of attack motives.
- **Limitations – time-consuming** and subjective **bias.**

TYPE OF DATA ANALYSIS (CONTINUES…)

ii. Behavioural

- Answers "**How did this happen?**".
- **Tool – Sandboxing**.
- **Strength** – detects unknown threats.
- **Limitations** – high false positives.

TYPE OF DATA ANALYSIS (CONTINUES…)

Factor	Qualitative	Quantitative
Data Type	Text, Logs and Narratives	Numbers, Metrics and Counts
Analysis Focus	Why and How?	What and How Much?
Tools	TheHive	Slunk and SIEM
Outcome	Threat Intelligence and root causes	Statistics

GAP ANALYSIS

UNIT 4

INTRODUCTION

- **Gap analysis** is a strategic tool used to **compare the current state** of a system, process, or performance **with the desired or target state**.
- It helps **identify discrepancies ("gaps")** and provides insights into necessary improvements.
- **Gap analysis helps organisations to –**

i. Identify vulnerabilities.

ii. Compare existing security measures with industry standards.

iii. Develop action plans to mitigate fraud risks.

STEPS IN CONDUCTING GAP ANALYSIS FOR CYBERSECURITY FRAUD PREVENTION

1. **Define Objectives** – Determine what fraud risks need to be **addressed** (e.g., identity theft).
2. **Assess Current State** (Current Cybersecurity Posture) – **Review** historical fraud incidents and response effectiveness.
3. **Identify Desired State – Benchmark** against industry standards (e.g., NIST guidelines).
4. **Identify Gaps – Compare** current vs. desired performance.
5. **Develop an Action Plan** – This can include **new detection techniques, authentication mechanisms** and **awareness.**

USING GAP ANALYSIS IN FRAUD DETECTION

1. **Fraud Detection Gaps** – include **Rule-Based Systems vs. AI/ML Models, Data Inconsistencies,** and **Response Delays.**
2. **Fraud Prevention Gaps – Weak Authentication** and **Lack of Employee Training.**

PERFORMING CONTEXTUAL AND BEHAVIOR QUALITATIVE ANALYSIS

UNIT 6

PREPARED BY JAMES SOKO FOR MR. MUKOMBO MUKOMBO © 2025 KAPASA MAKASA UNIVERSITY

KEY CONCEPTS IN CONTEXTUAL AND BEHAVIORAL ANALYSIS

1. **Keywords Analysis** – involves **scanning text data** (emails, chats, documents) for suspicious keywords that may indicate fraud.
2. **Content Analysis** (Files, Pictures, Videos) – **examines digital content** for signs of manipulation or malicious intent, **techniques** include **Metadata Analysis** and **Image Forensics**.
3. **Behavior Profiling** – builds a baseline of normal behavior for users/systems and flags deviations.

ONLINE PRESENCE & SOCIAL MEDIA ANALYSIS

1. **Social Media Investigations** – it involves **Identity Verification** and **Fraud Detection** (for any form of impersonation).
2. **Open-Source Intelligence (OSINT)** – gathers **publicly available data** (social media, forums, news) for fraud investigations.
3. **Deep & Dark Web Monitoring** – fraudsters sell **stolen data** on dark web markets, monitoring helps prevent breaches.

DIGITAL FORENSIC PROCESS

1. **Identification – locate** relevant data (emails, logs, hard drives).
2. **Preservation – secure** data to prevent tampering (hash verification).
3. **Analysis – examine** data for fraud indicators (deleted files, hidden folders).
4. **Reporting – document** findings for legal proceedings.

COMMON FORENSIC TECHNIQUES

1. **File Carving** – recovers deleted files.
2. **Memory Dump Analysis** – detects malware.
3. **Network Forensics** – analyzes traffic logs for unauthorized access.

Challenges in conducting Qualitative Fraud Analysis (keyword: **PEF**) –

i. Privacy Concerns
ii. Evolving Tactics
iii. False Positives

OLAF APPROACH IN DATA ANALYSIS

UNIT 7

INTRODUCTION

- The **OLAF** Approach (**Object, Locate, Analyze, Follow**) is a structured methodology used in **fraud investigations** and **data analysis** to systematically uncover fraudulent activities.
- Breakdown of the OLAF Approach

1. **Object (Identify the Fraud Object)**

- The "**Object**" refers to the **subject** of investigation (what is being examined for potential fraud).
- For example, a **fake social media account** used for scams.

BREAKDOWN OF THE OLAF APPROACH

2. Locate (Find Relevant Data Sources)

- Once an object is identified, investigators must **locate all related data** that could reveal fraud.
- **Data sources** include **digital footprints** and **external databases** (public records).
- **Techniques for locating data** include **Forensic Imaging** and **Open-Source Intelligence (OSINT).**

BREAKDOWN OF THE OLAF APPROACH (CONTINUES…)

3. **Analyze (Examine Evidence for Fraud Indicators)**

- In-depth **examination** of collected data to confirm fraud.
- Key **analysis methods** include **Behavioral Analysis** and **Digital Forensics.**

4. **Follow (Pursue Leads to Uncover Full Fraud Scheme)**

- Expanding the investigation to **track all connected parties** and fraud mechanisms.
- Steps involved include **Link Analysis, Surveillance & Monitoring** and **Legal Action.**

MISCARRIAGE OF JUSTICE

- This is when an **innocent person is wrongly convicted or punished** due to flawed fraud investigations.
- **Causes** in fraud cases include **False Positives, Biased Analysis** and **Tampered Evidence.**
- **Prevention measures** include **Cross-Verification** and **avoid Human Oversight** through the use of **AI.**

IMPURITY OF FINDINGS

- This is when **investigation results are contaminated** due to **Analyst Bias** and **Evidence Tampering**.
- To ensure **pure findings**, an investigator must document a **Chain of Custody**.
- **Challenges in the OLAF Approach** include –

i. Encryption & Privacy Laws

ii. **Data Overload** slows analysis